

1. What is a Network Interface Card?

Network Interface Card provides the network connectivity to the device or machine, modern machine has an Ethernet NIC, a wireless NIC, and virtual adapter.

2. What is Ethernet?

Ethernet is the technology works on Land Area Network, it generally provides speed of 10 mbps, 100 mps, 1 gbps, 10 gbps as well, and it all depends on the configuration.

3. What is RJ-45?

RJ-45 is an Ethernet cable with 8 positioned wires, it connects devices like routers, PCs, switches, etc., on LAN.

4. What does UTP stand for?

Unshielded Twisted Pair

5. Name three common Ethernet cable categories.

CAT5e, CAT6, CAT6a

6. What is a straight-through cable?

A straight-through cable is a traditional way to connect different types of devices with switch as a medium.

7. What is a crossover cable?

A crossover cable is used to connect similar devices with no switch medium.

8. Explain Auto MDI-X.

Auto MDI-X is a hardware feature in Ethernet ports that automatically detects whether a connected cable requires a straight-through or crossover configuration and internally swaps the transmission process.

9. Difference between copper and fiber-optic networking?

Copper networking is used at a short distance networking connectivity with limitations of distance and electromagnetic interference; While fibre-optic networking works with longer distance networking and immune to EMI.

10. What is multimode fiber?

It is a shorter-distance fiber cable and cheaper than single-mode cable, and does send every type of signal.

11. What is single-mode fiber?

It is a longer-distance fiber cable and is expensive than multimode cable, it is a laser data transmission cable.

12. Two advantages of fiber-optic networking.

- a) It used as a longer-distance networking connectivity.
- b) It is secured from data leakage because of the protection layers and immune to Electronic Magnetic Interference

13. Why is understanding network interfaces important for cybersecurity?

This knowledge is used for the investigation. And it is the foundation of Wireshark, Nmap, network security, penetration testing.

PRACTICAL IDENTIFICATION (4 marks)

14. Identify your active network adapter.

wlan0,

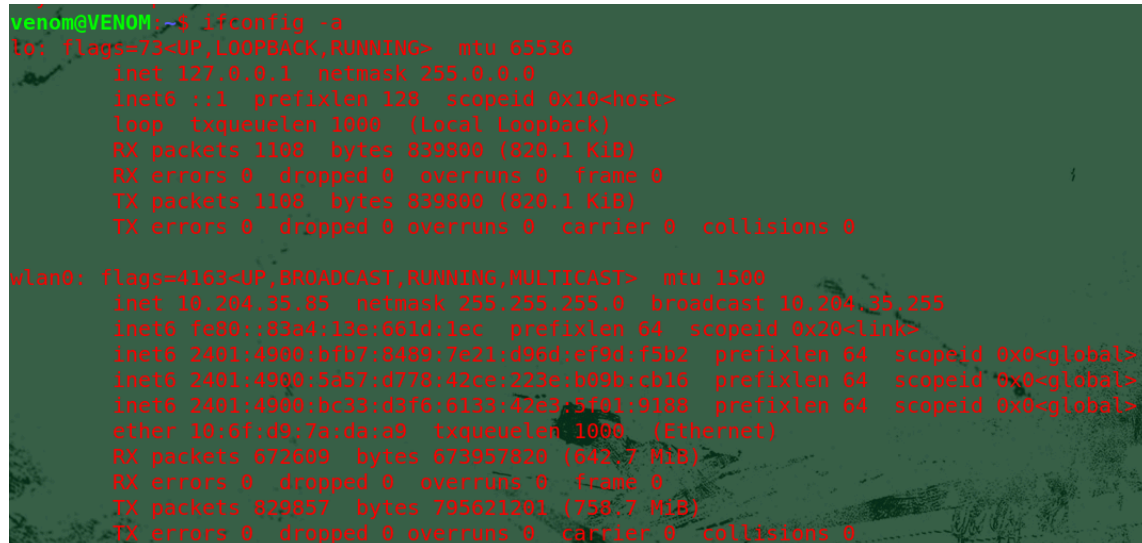
I have wireless network interface, not any physical adapter.

15. Write your IPv4 address, default gateway, and network adapter name.

IP : 10.204.35.85 , Gateway : 10.204.35.18 , Network Adapter : wlan0

SCREENSHOT (3 marks)

Attach a screenshot of: ipconfig /all



```
venom@VENOM:~$ ifconfig -a
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1100 bytes 839800 (820.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1100 bytes 839800 (820.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.204.35.85 netmask 255.255.255.0 broadcast 10.204.35.255
    inet6 fe80::63a4:13e:661d:1ec prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:bfb7:8489:7e21:d96d:ef9d:f5b2 prefixlen 64 scopeid 0x0<global>
    inet6 2401:4900:5a57:d778:42ce:223e:b09b:cb10 prefixlen 64 scopeid 0x0<global>
    inet6 2401:4900:bc33:d3f6:6133:42e3:5f61:9180 prefixlen 64 scopeid 0x0<global>
    ether 10:6f:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 672609 bytes 673957820 (642.7 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 829857 bytes 795621201 (758.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```